

Q.11

S = "a b c a"

→ Max ek character delete karke check karo ki string palindrome ban sakti hai ya nahi.  
→ Jo ulta aur seedha same ho.

Here,

S = a b c a  
↑ ↑ ↑  
left right

~~Ab kare~~ left = right.

left++.

right--.

∵ b != c ⇒ Ab hum ek character delete karne ki permission hai.

• b hatao → "a c a" → True

• c hatao → "a b a" → True.

→ To sabse pehle hm do pointer use karenge left and right.

int left = 0;

int right = s.length - 1.

Phir loop use karenge

while (left < right)

if (s.charAt(left) == s.charAt(right))

left++;

right--;

$\neg$   $a \quad b \quad c \quad a \quad \Rightarrow a = a$   
 $\uparrow \quad \quad \quad \uparrow$   
 left right.

$a \quad b \quad c \quad a$   
 $\uparrow \quad \uparrow$   
 left right.  $\Rightarrow b \neq c$

Ab dono me se ek character j skip karo.

① agar left ko skip karna hai to

$\boxed{\text{isPalindrome}(s, \text{left}+1, \text{right})}$

$\hookrightarrow$  Matlab c ko rehne do b ko skip karo.

② agar right ko skip karna hai to.

$\boxed{\text{isPalindrome}(s, \text{left}, \text{right}-1)}$

$\hookrightarrow$  Matlab b ko rehne do c ko skip karo.



// code -

```
class Soln {
```

```
    public void isPalindrome (string s) {
```

```
        int left = 0;
```

```
        int right = s.length - 1;
```

```
        while (left < right) {
```

```
            if (s.charAt(left) == s.charAt(right)) {
```

```
                left++;
```

```
                right--;
```

```
            } else {
```

```
                return isPalindrome (s, left+1, right-1);
```

```
                return isPalindrome (s, left, right+1);
```

```
            }
```

```
        }
```

```
        return True;
```

```
    }
```

```
    public boolean isPalindrome (string s, int left,  
                                int right) {
```

```
        while (left < right) {
```

```
            if (s.charAt(left) != s.charAt(right)) {
```

```
                return false;
```

```
            }
```

```
            left++;
```

```
            right--;
```

```
        }
```

```
        return True;
```

```
    }
```

```
}
```